

SIDONPAY

Fraud Detection & Transaction Analysis

PaySim synthetic Mobile Money Dataset --- week 3

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1. Introduction

This week's task moved from business immersion into hands-on technical work: analyzing SIDONPAY's future transaction data before the platform even launches. We used PaySim, a synthetic mobile-money dataset built from one month of real transaction logs from an African mobile financial service — 6,362,620 transactions across five types (CASH-IN, CASH-OUT, DEBIT, PAYMENT, TRANSFER), with 8,213 confirmed fraud cases.

2. Objective

This analysis examines transaction behavior, identifies fraud patterns, evaluates the existing isFlaggedFraud mechanism, and translates the findings into practical recommendations for SIDONPAY. The Week 3 brief requires data cleaning, analysis of transaction types and behavior, fraud detection analysis, evaluation of the current flagging system, and business recommendations.

3. Data Preparation & Methodology

The dataset was loaded and examined in Python. The workflow included checking structure, data types, missing values and duplicates, followed by descriptive and comparative analysis. The step field was used for time analysis because one step represents one hour and 744 steps represent the 30-day simulation. Fraud analysis used isFraud as the actual outcome and isFlaggedFraud as the existing system flag. Transaction amounts were grouped into value bands. Detection performance was evaluated using TP, FP, FN and TN, then precision, recall and accuracy were calculated.

Data limitation: the supplied PaySim documentation states that oldbalanceOrg, newbalanceOrig, oldbalanceDest and newbalanceDest should not be used as fraud-detection predictors because fraudulent transactions are cancelled in the simulation

4. Key Results

Metric	Result
Total Transactions	6,362,620
Fraud Transactions	8,213 (0.13%)
Fraud Amount	12.06 billion
Existing Flags	16
True Positives	16
False Positives	0
False Negatives	8,197
True Negatives	6354407
Precision	100%
Recall	0.19%
Accuracy	99.87%

Transaction pattern: all observed fraud occurred in TRANSFER and CASH_OUT transactions: 4,097 fraudulent TRANSFER transactions and 4,116 fraudulent CASH_OUT transactions. PAYMENT, CASH-IN and DEBIT recorded no fraud cases in this dataset.

Amount pattern: higher-value transactions showed greater observed fraud risk. The 1m+ band had the highest observed fraud rate at approximately 2.07%, compared with the overall rate of 0.13%. This supports using transaction value as a risk signal rather than treating a particular amount as automatic fraud.

5. Existing Fraud-Flagging System

The current mechanism flagged only 16 transactions. All 16 were actual fraud, producing 100% precision and zero false positives. However, 8,197 of the 8,213 actual fraud transactions were missed, producing only 0.19% recall. Although overall accuracy is 99.87%, that figure is misleading for fraud detection because fraud represents a very small share of all transactions. Recall and missed-fraud volume are more useful indicators of detection coverage.

6. Recommendations for SIDONPAY

- Prioritize TRANSFER and CASH_OUT monitoring because these accounted for all observed fraud.
- Use transaction amount as a risk signal, particularly for unusually large transactions, without using an amount-only rule.
- Move beyond the current narrow rule because its 0.19% recall shows that most fraud is missed.
- Develop multi-signal monitoring using transaction type, amount, unusual user behavior, transaction sequence and existing alerts.
- Investigate transfer-to-cash-out behavior as a potentially important fraud pattern.

7. Dashboard & Conclusion

The Power BI deliverable is organized into five pages: Transaction Overview; Fraud Pattern Analysis; User & Transaction Behaviour; Flag Evaluation; and Recommendations. Python provides the technical calculations while Power BI communicates the findings.

Overall, the analysis demonstrates that SIDONPAY's current flagging approach has excellent precision but extremely poor fraud coverage. A broader, multi-signal monitoring approach is therefore recommended.